
A Robot Policy Designed as a Weight-Space Artifact Exact Tensor Structure in a Capable VLA

Rational conversion, reduced-Gram analysis, and exact weight access

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Abstract

Tensor decomposition rewrites a multidimensional table as a small set of low-rank factors. In a neural network, those factors can expose which combinations of inputs and outputs are encoded by the weights. Conventional nonlinear operations make such a table difficult to obtain directly. We introduce χ -VLA, a 20.1M-parameter vision-language-action policy containing only bilinear, linear, and rational learned operations. Its checkpoint therefore specifies an explicit rational tensor program rather than only an opaque executable function. A mechanical audit and local reconstructions verify that claim. We derive an exact weight-based decomposition of each softmax-free attention layer and reconstruct all 12 modules in three checkpoints with worst relative error 2.56×10^{-7} . An $O(d^2)$ reduced Gram avoids materializing the coefficient tensor. All 36 primary block-head tests favor the rank-128 weight-derived subspace over fixed random controls. The full appendix screen finds the effect in 83/96 pairs and identifies where the signal concentrates across depth. A convolutional instantiation of the same rational model family reaches $93.7\% \pm 0.8\%$ on LIBERO-Object, and its three-checkpoint ensemble reaches 96.2%. We therefore establish a capable model family with exact layer-wise structure derived from trained coefficients. We do not yet provide an exact end-to-end decomposition of the complete policy.

1 Introduction

Neural checkpoints are usually treated as opaque containers for functions. We can run them and collect their activations, but the weights alone rarely provide a readable map from input features to outputs. A post-hoc probe may reveal that an activation correlates with behavior, yet it does not show how the weights construct that feature.

A tensor is a multidimensional table, and tensor decomposition is the analogue of low-rank matrix factorization for such tables. When a network uses only additions, multiplications, and ratios of polynomials, its weights can be expanded into coefficient tensors whose entries record input interactions. Decomposition ranks simpler factors inside those coefficients. The intended benefit is an inspectable model artifact: candidate structure comes from the trained parameters. Softmax, ordinary activations, and square-root normalization obstruct this direct conversion in conventional networks. Bilinear language and vision models suggest that changing the primitives can recover it [3, 4]. Whether the resulting artifact can still control a robot is unresolved.

We study two linked questions:

Can the learned weights of a specialist VLA specify an explicit rational tensor program, and can exact weight analysis coexist with strong closed-loop control?

35 Our contributions are:

- 36 1. We construct and mechanically certify a complete VLA from tensor-compatible primitives.
 37 A rational normalization supplies input-specific scaling without leaving the rational function
 38 class.
- 39 2. We derive exact weight-based structure through each softmax-free attention layer and an
 40 $O(d^2)$ reduced-Gram calculation that avoids materializing its high-order coefficient tensor.
- 41 3. We test exact attention subspaces against fixed equal-rank random controls. The weight-
 42 derived subspaces win all 36 primary block-head comparisons.
- 43 4. We establish capability in a convolutional instantiation of the same rational family, reaching
 44 $93.7\% \pm 0.8\%$ over three LIBERO-Object seeds. A three-checkpoint ensemble reaches
 45 96.2% at three times the inference cost.

46 This paper does not claim current state of the art. It also does not claim that comparable mechanisms
 47 are impossible to obtain from conventional policies. The empirical question is whether the algebraic
 48 policy makes exact weight structure directly accessible.

49 2 A rational tensor robot policy

50 2.1 Tensor-compatible primitives

51 Each primitive replaces a familiar transformer component while keeping explicit coefficients. Lin-
 52 ear maps contribute first-order terms, while multiplying learned projections creates higher-order
 53 interactions that remain determined by the weights.

54 **Homogeneous coordinate.** We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative
 55 layer to represent bias, linear, and quadratic terms in one contraction.

56 **Bilinear feed-forward block.** For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

57 The inner sum is a table indexed by one output and two input coordinates. The three matrices give its
 58 CP factorization, the tensor analogue of a low-rank matrix factorization.

59 **Softmax-free bilinear attention.** Each head forms two query-key systems and one value stream:

$$Y = C[M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

60 Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key,
 61 value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with
 62 interaction coefficients determined by the weights.

63 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
 64 root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z + \epsilon}$:

$$\text{RNorm}(x) = x r \left(\frac{1}{d} \sum_{j=1}^d x_j^2 \right). \quad (3)$$

65 Here rational means a ratio of polynomials. Projective coordinates carry each activation as a
 66 numerator-denominator pair (N, D) representing N/D . Updating this pair preserves input-specific
 67 rescaling without leaving the rational tensor representation.

68 2.2 Architecture and training

69 The policy follows the high-level VLA sequence of visual encoding, language and state embedding,
 70 joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional
 71 robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width
 72 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven
 73 continuous action dimensions. The complete model has 20,137,352 learned parameters.

74 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
 75 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
 76 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
 77 that verify every requested state was loaded.

78 2.3 Mechanical certificate on the trained checkpoint

79 Architecture definitions can differ from deployed code. We therefore audit the real 189/200 check-
 80 point at runtime and reconstruct representative operations from saved weights.

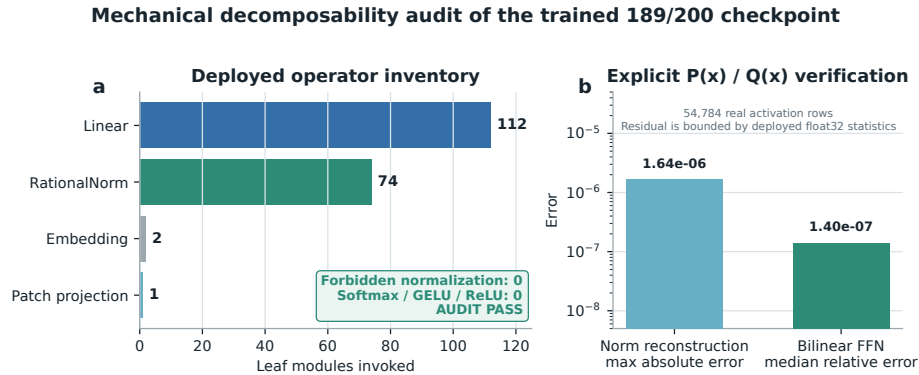


Figure 1: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.

81 The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN re-
 82 construction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed
 83 module’s float32 statistic. The audit certifies operator membership and local reconstruction. It does
 84 not materialize one compact polynomial for the full eight-layer transformer.

85 3 The artifact is a capable closed-loop policy

86 3.1 Evaluation protocol

87 We evaluate all ten LIBERO-Object tasks with:

- 88 • 50 canonical trials per task
- 89 • a 280-step cap
- 90 • three independently trained seeds per architecture
- 91 • 500 rollouts per seed and 1,500 rollouts per architecture

92 Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not
 93 rerun in our codebase, so differences may still include implementation and training choices.

94 3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational conv	20.1M	3	93.7% $\pm$ 0.8%
χ -VLA, conv ensemble	$3 \times 20.1\text{M}$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	88.4% $\pm$ 0.8%
Diffusion Policy [1]	not matched	3	92.5% $\pm$ 0.7%

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

95 The convolutional seeds score 94.0%, 94.4%, and 92.8%. Their narrow spread shows that high
 96 specialist capability is reproduced across three independently trained checkpoints. These are historical
 97 Modal aggregate records. A locked replay is producing immutable episode-level outcomes.

98 Separately, the paired ensemble evaluation re-evaluates the three members at 93.6%, 95.0%, and
 99 93.2%, for a 93.9% mean. Elementwise prediction averaging reaches 96.2%. This is 2.3 points
 100 above that paired mean and 1.2 points above its strongest member. The gain is largest on tomato
 101 sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one
 102 20.1M-parameter policy.

103 3.3 What this result does and does not establish

104 The result shows that the rational architecture supports high closed-loop specialist capability. It is an
 105 existence and reproducibility claim for a compact tensor-decomposable policy, not a claim of broad
 106 robotic generalization or state of the art.

107 4 Exact weight-derived structure through attention

108 4.1 Exact layerwise construction

109 Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives
 110 a coefficient table whose entries measure how combinations of query, key, and value coordinates
 111 contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-
 112 token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to
 113 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

114 The deployed attention also normalizes each query and key with Equation 3. Each normalization
 115 multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled
 116 outside the bilinear dot products. The polynomial attention core is unchanged, while explicit
 117 numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$
 118 10^{-16} . We then reconstruct all four vision and eight joint attention modules in each of three
 119 trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the
 120 worst absolute difference is 2.23×10^{-5} . This residual comes from the deployed implementation’s
 121 intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

122 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
 123 values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors
 124 rank important directions without constructing the full table. It matches brute-force materialization to
 125 2.13×10^{-14} at toy scale.

126 4.2 Trained policy result

127 For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve
 128 heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are
 129 sampled once, then fixed and reused for all examples. The weights choose the subspace, while cached
 130 activations are used only afterward to score how well it preserves the layer’s computation.

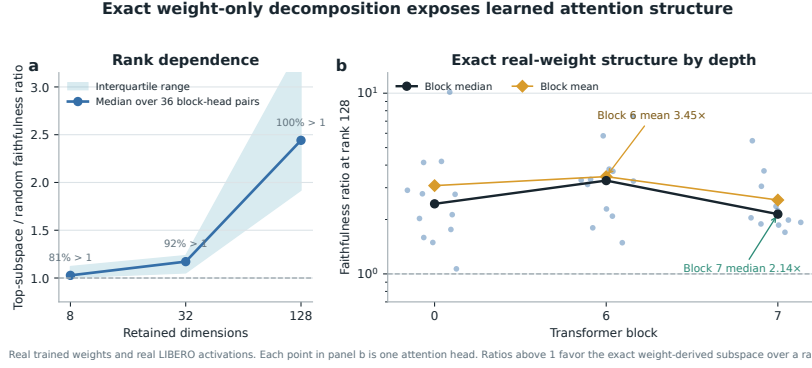


Figure 2: Exact weight-derived attention structure. Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

5 Related work and limitations

Related work. OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative weights can expose low-rank structure and support subsequent controls [3–5]. Recent work also steers pretrained VLAs through activation directions [6]. Our distinction is the exact weight-derived construction in a policy designed as a rational tensor program. Tensor and polynomial networks provide the broader multilinear foundation [12–14]. VLA robustness benchmarks show that standard LIBERO success can coexist with weak language use [9–11].

Limitations and conclusion. The benchmark is one specialist suite, so it does not establish broad robotic generalization. Exact reconstruction is certified layer by layer and does not yet produce one compact symbolic contraction of the full policy. The capability result uses convolutional checkpoints, while the exact analysis uses separate ViT checkpoints. Evaluation uses fixed task instructions and does not establish robust instruction grounding. Within that scope, rational restrictions support strong LIBERO-Object control and exact trained-weight certificates. We do not claim a direct coefficient-edit interface.

Reproducibility. The final artifact will provide code, configurations, checkpoint hashes, exact certificates, and per-episode logs. Historical Modal cache-backed analyses read dataset-native task metadata. The replacement Athena pipeline pins that metadata and maps dataset identifiers to official identifiers before defining task splits. Canonical-state checks fail loudly when requested states are unavailable.

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183 A Supporting results and decisions

184 A.1 Per-task ensemble behavior

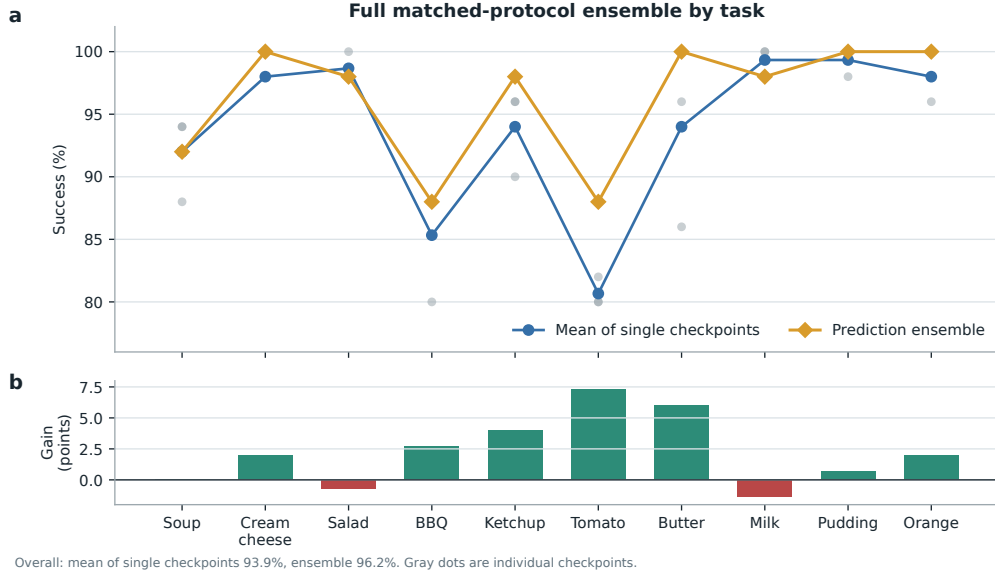


Figure 3: **Per-task effect of prediction averaging in the paired ensemble evaluation.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

185 A.2 Expanded attention screen

186 The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128,
 187 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean
 188 is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only 54/96 and 60/96
 189 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider
 190 tested rank.

191 A.3 Numerical scope of rational conversion

192 The deployed Padé computation is exactly rational even though it approximates RMS normalization.
 193 Final certification must report its polynomial degrees, coefficients, denominator margin, observed
 194 activation range, tail error, action drift, and runtime overhead. Operator membership alone does not
 195 establish practical numerical safety under distribution shift.